

STACK (Method Area)

main(String[] args)

Stack Frame

Variable	Value
args	reference to the command line
myCar	0x1001

myCar is a reference variable stored on the Stack which points to the Car object in Heap.

HEAP (Objects Area)

Object: Car (created by new Car())

Address: 0x1001

Field (Variable)	Value
color	"Black"
speed	0
engineOn	false
leftIndicatorOn	false
rightIndicatorOn	false

Object State During Execution

1. After new Car()

color	Black
speed	0
engineOn	false
leftInd	false
rightInd	false

2. After startEngine()

color	Black
speed	0
engineOn	true
leftInd	false
rightInd	false

3. After accelerate()

color	Black
speed	20
engineOn	true
leftInd	false
rightInd	false

4. After turnLeft()

color	Black
speed	20
engineOn	true
leftInd	true
rightInd	false

5. After brake()

color	Black
speed	10
engineOn	true
leftInd	true
rightInd	false

6. After indicatorsOff()

color	Black
speed	10
engineOn	true
leftInd	false
rightInd	false

7. After stopEngine()

color	Black
speed	0
engineOn	false
leftInd	false
rightInd	false

8. After show() (Final)

color	Black
speed	0
engineOn	false
leftInd	false
rightInd	false

Legend

0x1001 = Memory Address (Reference)

Flow Rules

- New object: Car() is created in Heap.
- Reference variable (myCar) is stored on Stack.
- All operations (methods) affect Heap object.
- Methods operate on the object via reference.